

Korenjski kriterij

$$\sum a_n ; a_n \geq 0$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} \begin{cases} < 1 & \text{konv.} \\ > 1 & \text{div.} \\ = 1 & \text{idk} \end{cases}$$

2.

a) $\sum \left(\frac{n-2}{n+2} \right)^{n(n+1)}$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{n-2}{n+2} \right)^{n(n+1)}} = \lim_{n \rightarrow \infty} \left(1 + \frac{-4}{n+2} \right)^{n+1} = e^{-4} < 1$$

$\Rightarrow$ je konv.

b)

$$\sum_{n=1}^{\infty} \frac{n^2}{\left(2 + \frac{1}{n} \right)^n}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n^2}}{\left(2 + \frac{1}{n} \right)} = \lim_{n \rightarrow \infty} \frac{n^{\frac{2}{n}}}{2 + \frac{1}{n}} = \frac{1}{2} < 1 \Rightarrow \text{konv.}$$

3. za katere $a > 0$ vrsta konv.?

$$\sum_{n=1}^{\infty} \frac{n!}{(a+1)(a+2)\dots(a+n)}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{(a+1)}\cancel{(a+2)}\dots\cancel{(a+n)}(n+1)!}{n! \cancel{(a+1)}\cancel{(a+2)}\dots\cancel{(a+n)}(a+n+1)} = \lim_{n \rightarrow \infty} \frac{n+1}{a+n+1} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{\frac{a}{n} + 1 + \frac{1}{n}} = 1$$

Rabejev test:

> 1 konv.

< 1 di.

$$\lim_{n \rightarrow \infty} u\left(\frac{a_n}{a_{n+1}} - 1\right) \begin{cases} < 1 & \text{allv.} \\ > 1 & \text{idk} \end{cases}$$

$$\lim_{n \rightarrow \infty} n \left(\frac{a+n+1}{n+1} - 1 \right) = \lim_{n \rightarrow \infty} \frac{a+n+1 - (n+1)}{n+1} =$$

$$= \lim_{n \rightarrow \infty} \frac{a_n}{n+1} = \lim_{n \rightarrow \infty} \frac{a}{1 + \frac{1}{n}} = a$$

↙
če $a > 1$: konvergira

$$a=1: \sum_{n=1}^{\infty} \frac{n!}{2 \cdot 3 \cdot \dots \cdot (n)(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n+1} > \infty \Rightarrow \text{div.}$$

8. 2 upoštevanjem $A = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$
 miži: $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

poišči vsake vrste:

$$B = 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \dots + \frac{1}{2n-1} - \frac{1}{4n-2} - \frac{1}{4n} + \dots$$

$$A: (-1)^{h-1} \left(\frac{1}{h} \right)$$

$$A = \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{2^n}$$

$$L1: \frac{1}{h} \xrightarrow{h \rightarrow 0} 0$$

L2: padajoče ✓

Konu. pagojno

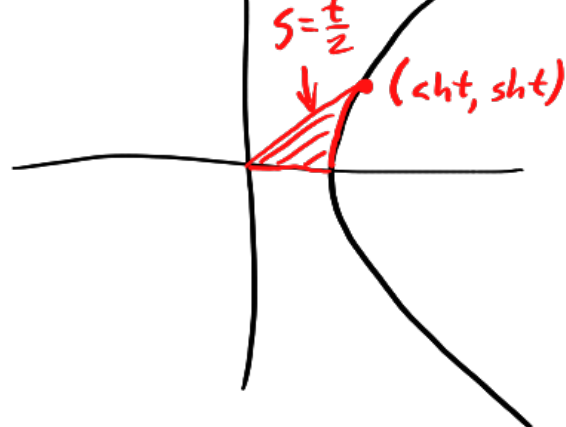
$$B = \sum \frac{1}{2^{n-1}} - \frac{1}{2} \left(\sum \frac{1}{2^{n-1}} + \sum \frac{1}{2^n} \right) = \frac{1}{2} \left(\sum \frac{1}{2^{n-1}} - \sum \frac{1}{2^n} \right) = \underline{\underline{\frac{1}{2} \ln 2}}$$

Feedback

$$1. \quad x^2 - y^2 = 1$$

1. $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = f(x)$

$D_f, Z_f, \text{graf}, f^{-1}(x)$



$D_f = \mathbb{R}$

$x = \frac{e^y - e^{-y}}{e^y + e^{-y}} = \frac{e^{2y} - 1}{e^{2y} + 1}$

$e^{2y} \cdot x - e^{2y} = -1 - x$

$e^{2y}(1-x) = x+1$

$e^{2y} = \frac{x+1}{1-x}$

$2y = \ln \frac{x+1}{1-x}$

$y = \frac{1}{2} \ln \frac{x+1}{1-x} = f^{-1}(x) = \text{arth } x$

$Z_f: \frac{x+1}{1-x} > 0$

① $x > 1:$

$x+1 < 0$

$x < -1$

NI REŠITEV

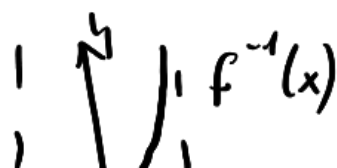
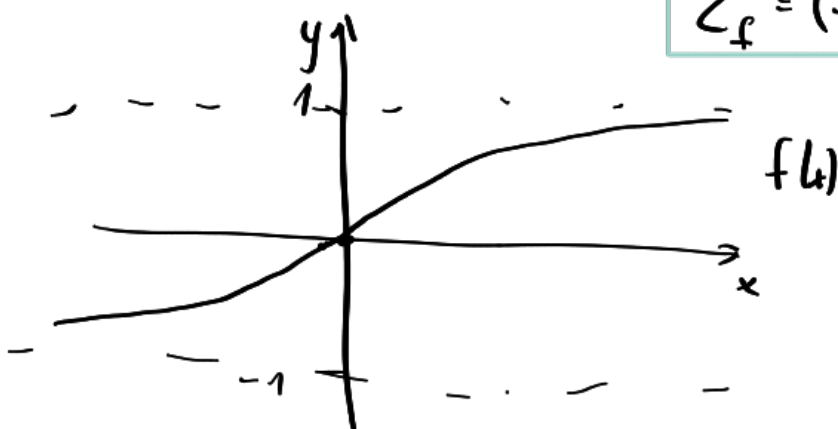
② $x < 1:$

$x+1 > 0$

$x > -1$

$-1 < x < 1$

$Z_f = (-1, 1)$





$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\tanh(x+y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$$

2.

$$a) \lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + 3x + 2} = \lim_{x \rightarrow -1} \frac{(x+1)(x-1)}{(x+1)(x+2)} = \frac{-2}{1} = -2$$

$$b) \lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right) = \lim_{x \rightarrow 1} \frac{1-x^3-3+3x}{(1-x)(1-x^3)} =$$

$$= \lim_{x \rightarrow 1} \frac{-x^3+3x-2}{(1-x)(1-x^3)} = \lim_{x \rightarrow 1} \frac{(x-1)(-x^2-x+2)(-1)}{(1-x)(1-x^3)} = \lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{1-x^3} =$$

-1	0	3	-2
1	-1	-1	2
-1	-1	2	0

$$= \lim_{x \rightarrow 1} \frac{(x+2)(x-1)(-1)}{(1-x)(1+x+x^2)} = \frac{(-1)(3)}{3} = -1$$

$$c) \lim_{x \rightarrow \infty} (\sqrt{x^2 - 5x + 6} - x) = \lim_{x \rightarrow \infty} \frac{x^2 - 5x + 6 - x^2}{\sqrt{x^2 - 5x + 6} + x} = \lim_{x \rightarrow \infty} \frac{-5 + \frac{6}{x}}{\sqrt{1 - \frac{5}{x} + \frac{6}{x^2}} + 1} =$$

$$= \frac{-5}{2}$$

$$d) \lim_{x \rightarrow \infty} \frac{x + \sin x}{x - \sin x} = \lim_{x \rightarrow \infty} \left(1 + \frac{2 \sin x}{x - \sin x} \right) = 1 + \lim_{x \rightarrow \infty} \frac{2 \sin x}{x - \sin x}$$

